

Disaster Impact Assessment

Group 2

Team Members

- Abdelrhman Y. Soliman
- Sergio Esteban Romero
- Javier Rodríguez Sánchez
- Javier Usón Peirón

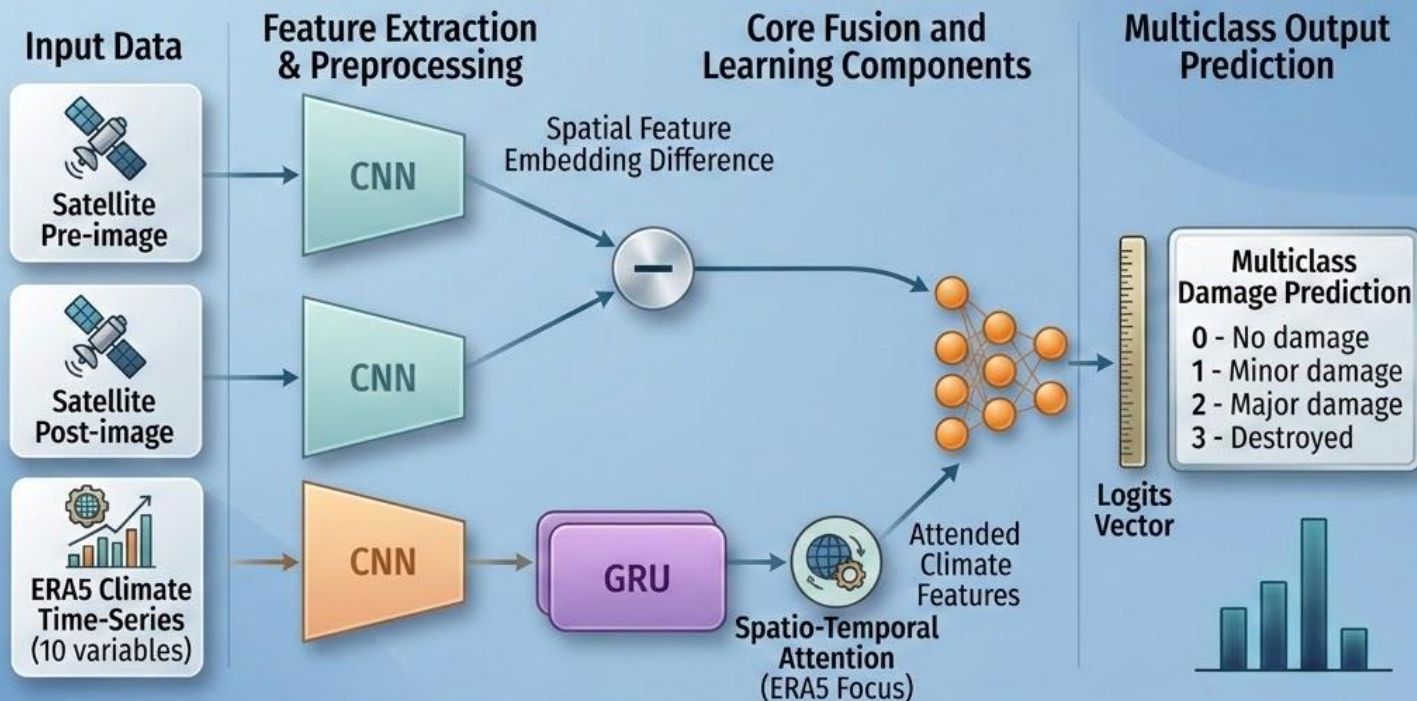
Group 2



UNIVERSIDAD
POLITÉCNICA
DE MADRID

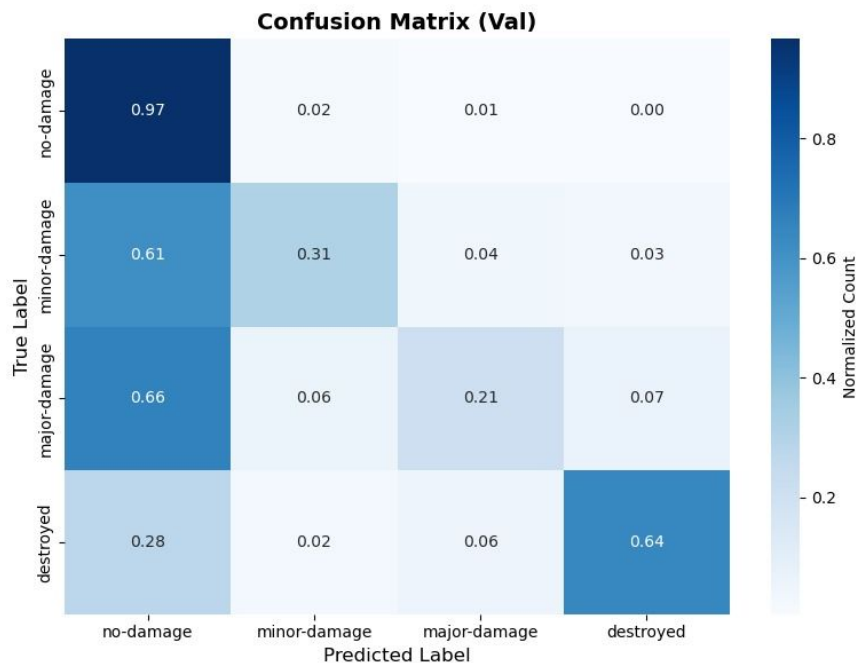


Network Architecture

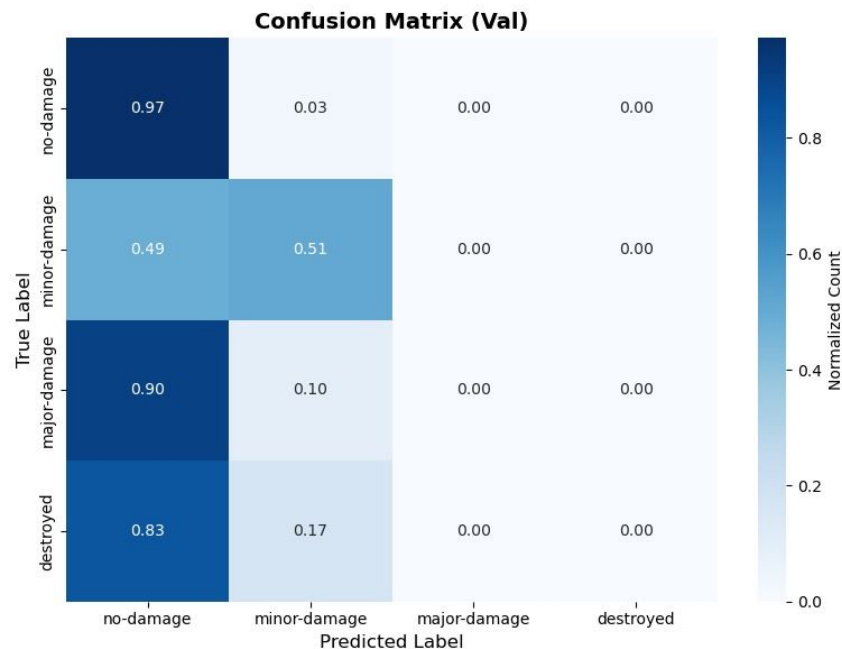


Evaluation Metrics (CM)

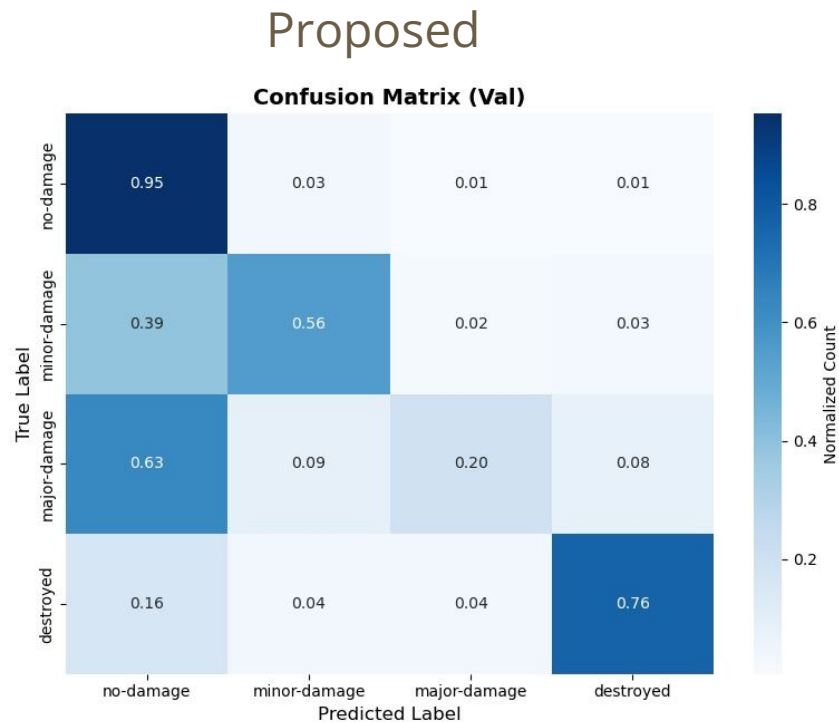
Images only



Climate only



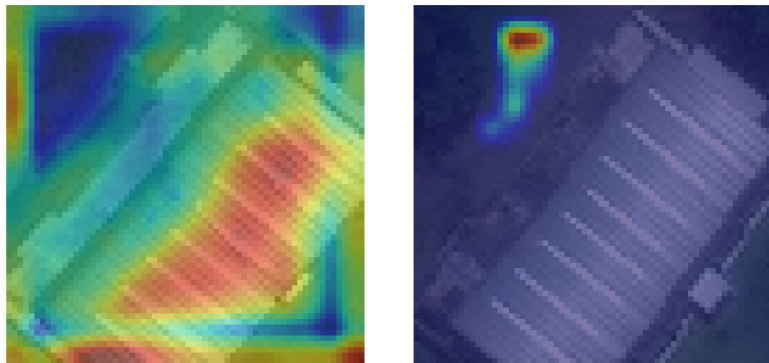
Evaluation Metrics (CM)



XAI and Interpretability - Grad CAM

No Damage

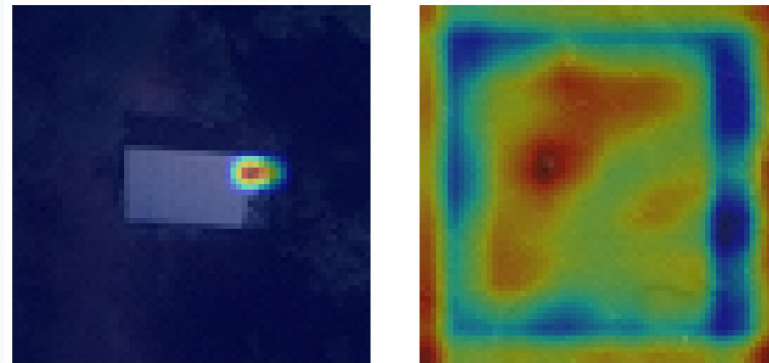
guatemala-volcano_00000009_pre_disaster. — GT: no-damage Pred: no-damage
Pre-Disaster Grad-CAM Post-Disaster Grad-CAM



Explainable AI visualization showing Grad-CAM heatmaps for pre and post-disaster satellite imagery classified as 'No Damage'.

Destroyed

guatemala-volcano_00000003_pre_disaster. — GT: destroyed Pred: destroyed
Pre-Disaster Grad-CAM Post-Disaster Grad-CAM

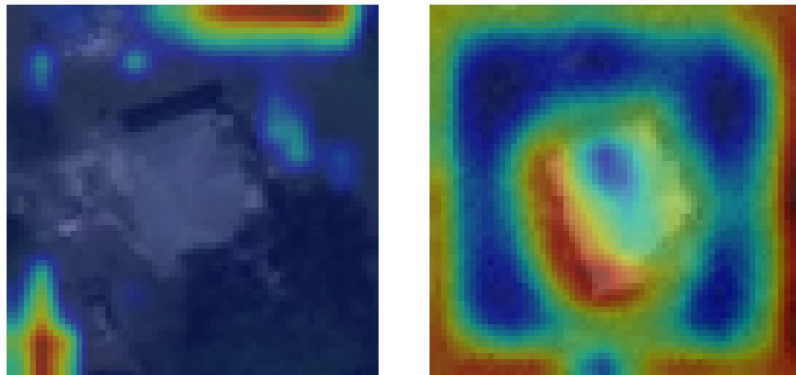


Visualization highlighting significant changes in the post-disaster Grad-CAM activations for areas classified as 'Destroyed'.

XAI and Interpretability - Grad CAM

Major Damage

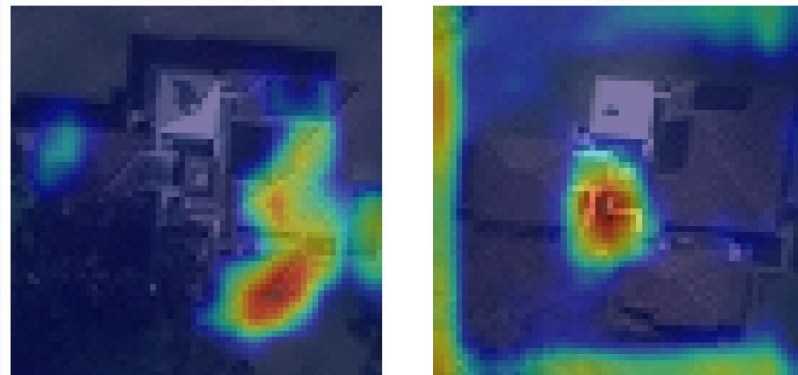
hurricane-florence_00000029_pre_disaster. — GT: major-damage Pred: major-damage
Pre-Disaster Grad-CAM Post-Disaster Grad-CAM



Explainable AI visualization showing Grad-CAM heatmaps for pre and post-disaster satellite imagery classified as 'Major Damage'.

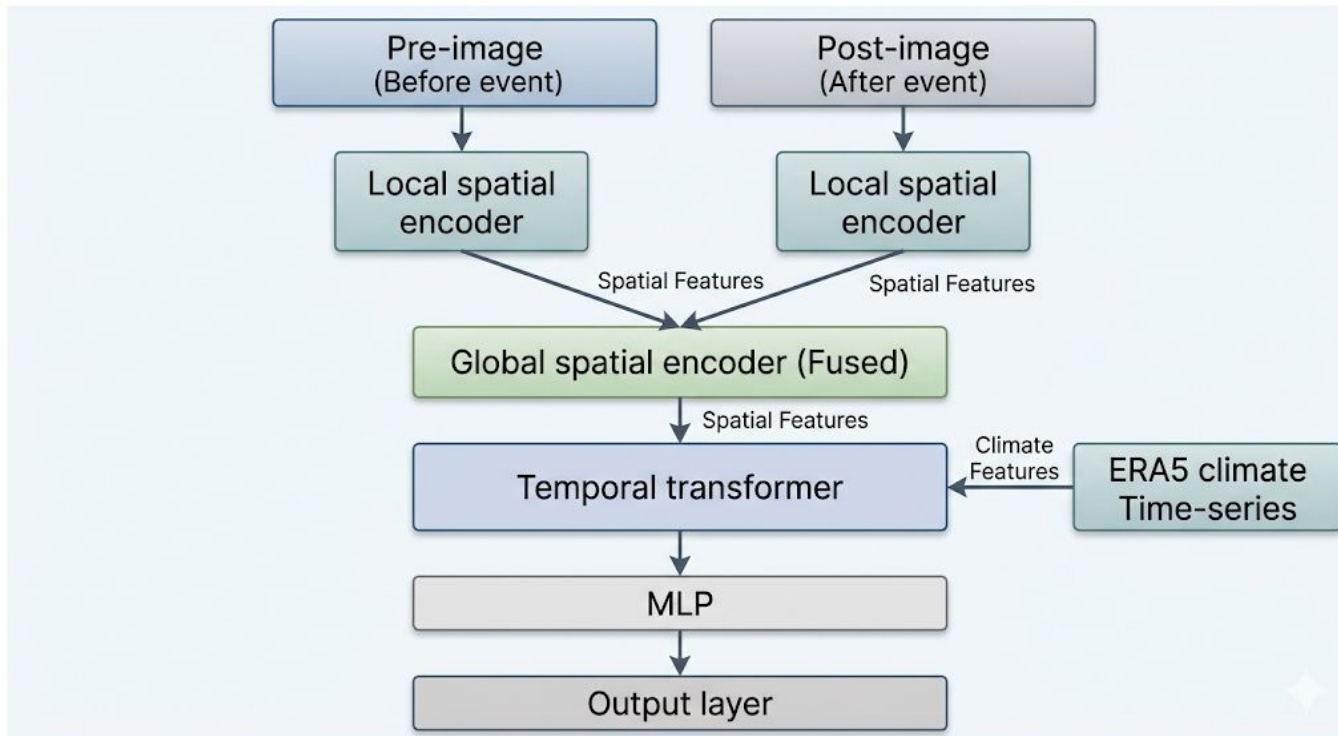
Minor Damage

guatemala-volcano_00000003_pre_disaster. — GT: minor-damage Pred: no-damage
Pre-Disaster Grad-CAM Post-Disaster Grad-CAM



Visualization highlighting significant changes in the post-disaster Grad-CAM activations for areas labeled 'Minor damage'.

Future Work





Thank You!

Any Questions?